

Global HIV Burden & Treatment Coverage Analysis

Insights on Global HIV Trends and Treatment Coverage

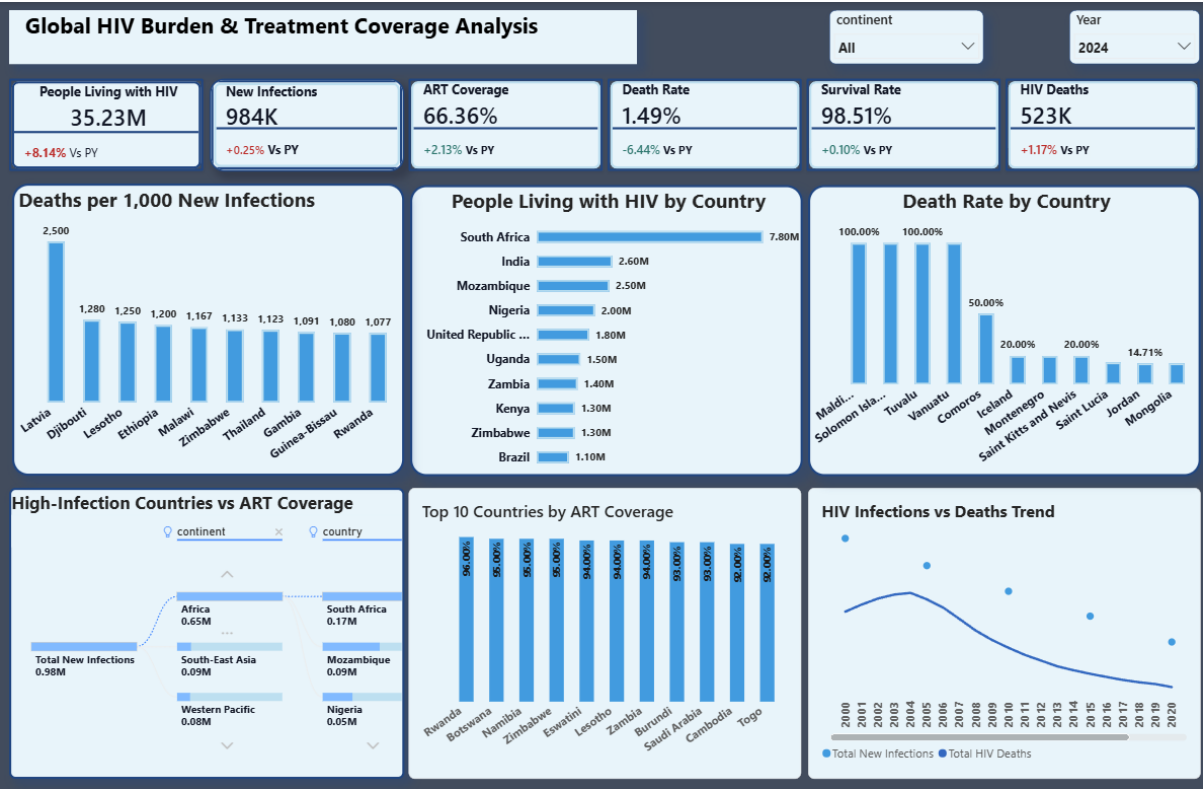
Mariam Adetoyi

Project Overview

HIV remains a major global public health concern despite progress in awareness, prevention, testing, and treatment. The expansion of Antiretroviral Therapy (ART) has improved survival outcomes, but new infections, HIV-related deaths, and unequal treatment access still remain challenges.

This project analyses global HIV data from **2000 to 2024** using **Power BI**. The dataset was cleaned, modelled into a **star schema**, and transformed into an interactive dashboard.

The dashboard allows users to explore HIV infections, deaths, ART coverage, mortality rate, survival rate, and country-level risk by **year**, **continent**, and **country**. The main dashboard focuses on **2024** to show the most recent global HIV status, while the trend analysis compares infections and deaths over time.



Business Objectives

The analysis was designed to answer the following key public health questions:

- How many new HIV infections are occurring over time?
- What is the total number of HIV-related deaths globally?
- How many people are currently living with HIV?
- What is the overall level of ART treatment coverage across countries?
- What are the mortality rate and survival rate of the HIV population?
- For every 1,000 new HIV infections, which countries record the highest number of deaths?
- How does the trend of new infections compare with HIV-related deaths?
- Which countries and regions are most affected by HIV?
- Which countries have the best treatment access?
- Do high-infection countries have adequate treatment coverage?
- Does higher ART coverage appear to support lower mortality?
- Which countries are most at risk due to poor treatment access or high mortality?

These questions guided the dashboard structure, KPI selection, DAX measures, and visual storytelling approach.

Data Preparation & Modelling

The raw HIV dataset was imported into Power BI and transformed using Power Query. The dataset contained annual records by country, year, indicator, and value.

The key fields in the dataset included:

- Country: The geographic location where the HIV data was recorded.
- Year: The annual time period of each record.

- Indicator: The HIV metric being measured.
- Value: The numeric value attached to each indicator.
- The main indicators analysed were:
- New infections: Number of newly reported HIV infections.
- HIV deaths: Number of deaths caused by HIV-related conditions.
- People living with HIV: Total number of people currently living with HIV.
- ART coverage: Percentage of HIV-positive individuals receiving treatment.

To improve the quality and structure of the analysis, the dataset was modelled using a star schema approach. This helped separate the fact table from descriptive dimension tables and made the dashboard easier to analyse.

The final data model included:

- FactHIV: The main fact table containing HIV values.
- DimDate: Used for year-based analysis and YoY comparison.
- DimLocation: Used for country and continent-level analysis.
- DimIndicator: Used to identify each HIV metric.
- Measure Table: Used to organise DAX measures such as death rate, survival rate, ART coverage, HIV deaths, new infections, and YoY growth.

This structure created a reliable foundation for KPI development, trend analysis, and interactive dashboard filtering.

Dashboard Development

The final dashboard was designed as a clean, one-page Power BI report with a healthcare-inspired visual style. The design used a dark background, light blue visual containers, blue chart accents, and clear KPI cards.

The dashboard includes interactive slicers for:

- Year
- Continent

The main dashboard features include:

KPI cards for:

- People living with HIV
- New infections
- ART coverage
- Death rate
- Survival rate
- HIV deaths.

YoY comparison beneath each KPI card.

- Deaths per 1,000 new infections by country.
- People living with HIV by country.
- Death rate by country.
- High-infection countries vs ART coverage using a decomposition tree.
- Top 10 countries by ART coverage.
- Trend chart comparing new infections and HIV deaths over time.

The dashboard was filtered to 2024 for the main view because this provides the latest available global HIV position. This avoids overstating current values by summing annual records across multiple years, especially for the “people living with HIV” metric.

Key KPI Insights

The 2024 dashboard provides the most recent view of global HIV status.

In **2024**, the dashboard showed:

- **35.23 million** people living with HIV.

- **984 thousand** new HIV infections.
- **523 thousand** HIV-related deaths.
- **66.36%** ART coverage.
- **1.49%** death rate.
- **98.51%** survival rate.

These results show that HIV remains a major global health burden. Although survival outcomes are strong, millions of people still live with HIV and almost one million new infections were recorded in 2024.

The KPI cards also included YoY indicators. Compared with the previous year:

- People living with HIV increased by **8.14%**.
- New infections increased slightly by **0.25%**.
- ART coverage improved by **2.13%**.
- Death rate reduced by **6.44%**.
- Survival rate improved by **0.10%**.
- HIV deaths increased by **1.17%**.

The improvement in ART coverage and reduction in death rate suggest better treatment access and disease management. However, the increase in new infections and HIV deaths shows that HIV still requires continuous intervention.

Trend Analysis

The trend analysis compared new HIV infections and HIV-related deaths between **2000 and 2024**.

This analysis showed that:

- Total new infections decreased by **59.02%** between 2000 and 2024.
- Total HIV deaths decreased by **65.50%** over the same period.
- New infections began a stronger downward trend from **2015**, falling by **479,090 cases** over nine years.
- HIV deaths recorded their steepest decline between **2009 and 2012**, dropping from **1,174,630 to 926,320**.

This shows that global HIV outcomes have improved significantly over time. The decline in deaths suggests that treatment access, healthcare management, and ART coverage have likely contributed to better survival outcomes.

However, the 2024 figures show that HIV is still a major public health challenge, with **984K new infections** and **523K HIV-related deaths** still recorded.

Country-Level Analysis

The country-level visuals show that HIV burden is not evenly distributed across countries.

In 2024, **South Africa** recorded the highest number of people living with HIV, with approximately **7.80 million** people affected.

Other high-burden countries included:

- India
- Mozambique
- Nigeria
- Uganda
- Zambia
- Kenya
- Zimbabwe
- Brazil

This shows that the HIV burden remains concentrated in specific countries, especially countries in Africa.

The all-year Power BI narrative also showed that **South Africa accounted for 33.50% of total people living with HIV** in the selected historical view. This confirms South Africa as one of the most significant contributors to HIV burden in the dataset.

Mortality and Risk Analysis

The dashboard included both **HIV deaths** and **death rate** because they explain different things.

HIV deaths show the actual number of HIV-related deaths.

Death rate shows the proportion of people living with HIV who died from HIV-related causes.

In 2024, the HIV death rate was **1.49%**, while the survival rate was **98.51%**. This indicates that most people living with HIV survived during the selected year.

The dashboard also included a “Deaths per 1,000 New Infections” visual. This metric helps identify countries where mortality pressure appears high compared with new infection levels.

In the 2024 dashboard, countries such as Latvia, Djibouti, Lesotho, Ethiopia, Malawi, Zimbabwe, Thailand, Ghana, Guinea-Bissau, and Rwanda appeared among countries with high deaths per 1,000 new infections.

This metric should be interpreted carefully. It does not mean newly infected people died immediately. Instead, it compares HIV deaths in the selected year against new infections in the same year. It is useful as a risk indicator, but not as a direct new-case death rate.

Treatment Coverage Analysis

ART coverage was included as a key treatment access indicator. In 2024, the dashboard showed **66.36% ART coverage**, meaning about two-thirds of people living with HIV were receiving treatment.

The top countries by ART coverage in the 2024 view included:

- Rwanda
- Botswana
- Namibia
- Zimbabwe
- Eswatini
- Lesotho
- Zambia
- Burundi
- Saudi Arabia

- Cambodia
- Togo

These countries showed stronger treatment access compared with others in the dataset.

The decomposition tree answer the question:

Do high-infection countries have adequate treatment coverage?

The analysis showed that high-infection countries do not always have the same level of treatment access. In 2024, Africa had the highest number of new infections, with South Africa, Mozambique, and Nigeria appearing as high-infection countries.

This means that treatment coverage must remain a priority in high-burden countries. Even where treatment access is improving, countries with high infections need continued prevention, early diagnosis, and long-term ART support.

Recommendations

Based on the analysis, the following recommendations are proposed:

1. Expand ART Coverage in High-Burden Countries

Countries with large HIV populations should continue expanding ART coverage. Treatment access is essential for reducing mortality and improving survival outcomes.

2. Strengthen HIV Prevention Programmes

Although new infections have declined over time, the 2024 dashboard still shows **984K new infections**. This means prevention campaigns, testing, education, and awareness programmes remain important.

3. Prioritise Countries with High Death Rates

Countries with high mortality rates should be investigated further to understand whether the issue is caused by poor treatment access, late diagnosis, weak healthcare systems, or reporting problems.

5. Focus on Africa and Other High-Burden Regions

Africa remains a major contributor to new HIV infections and overall HIV burden. Continued investment in prevention, ART access, testing, and long-term care is needed.

6. Monitor YoY Changes Annually

Public health stakeholders should track new infections, HIV deaths, ART coverage, death rate, and survival rate each year to identify progress or emerging risks.

7. Use Power BI for Continuous Monitoring

The dashboard should be updated regularly as new data becomes available. This will help decision-makers monitor HIV trends and allocate resources more effectively.

Conclusion

This project demonstrates how Power BI can be used to transform global HIV data into actionable public health insights.

The analysis shows that global HIV outcomes improved between **2000 and 2024**, with total new infections declining by **59.02%** and HIV deaths declining by **65.50%**. In 2024, the dashboard showed **35.23 million people living with HIV, 984K new infections, 523K HIV deaths, 66.36% ART coverage, 1.49% death rate, and 98.51% survival rate.**

These results suggest that ART coverage and improved HIV management have contributed to better survival outcomes. However, HIV remains a serious global health challenge because new infections continue to occur, deaths remain significant, and treatment access is not equal across countries.

Overall, the dashboard provides a clear and interactive tool for monitoring HIV burden, identifying high-risk countries, evaluating treatment coverage, and supporting evidence-based public health decision-making.